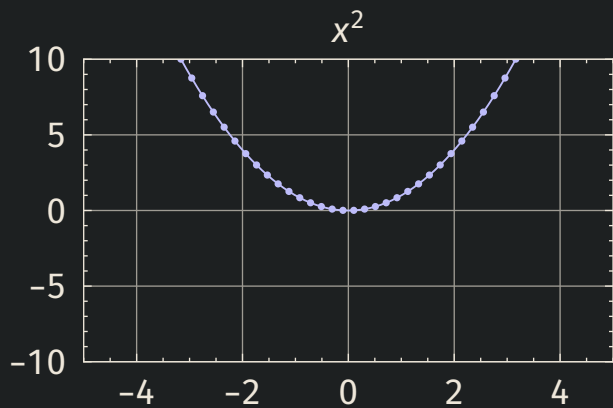


Functions

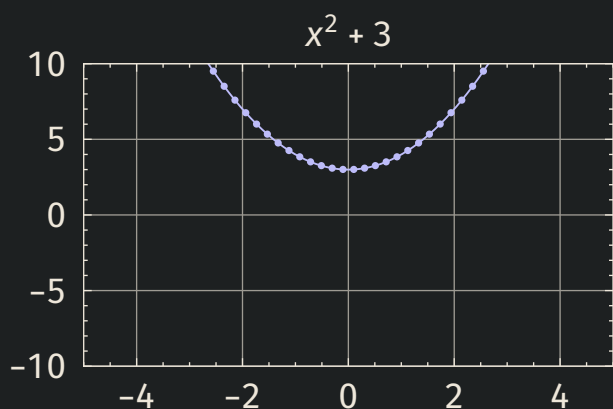
1. Transformation of functions

1.1. Translations

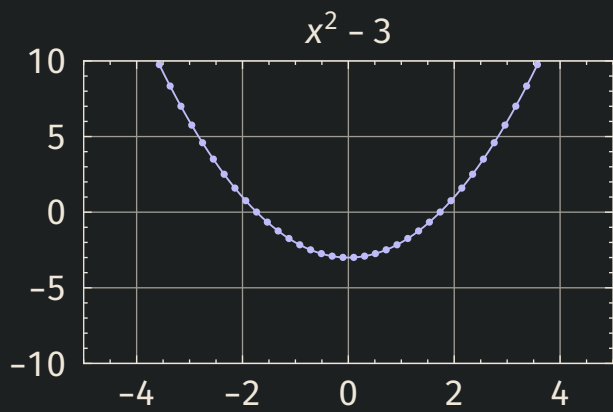
- $f(x) = x^2 \rightarrow$ Parent Function



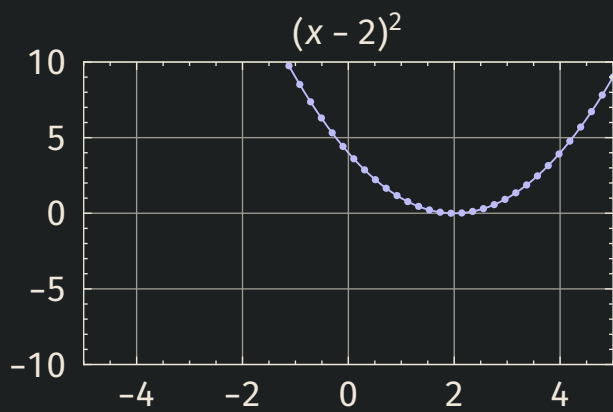
- $f(x) + c \rightarrow$ Shifts the graph c units up:



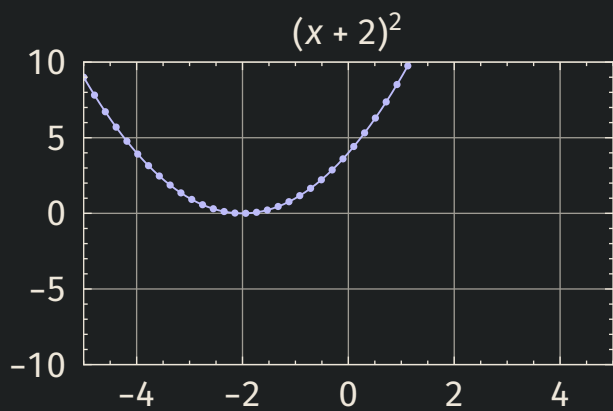
- $f(x) - c \rightarrow$ Shifts the graph c units down



- $f(x - c) \rightarrow$ Shifts the graph c units right

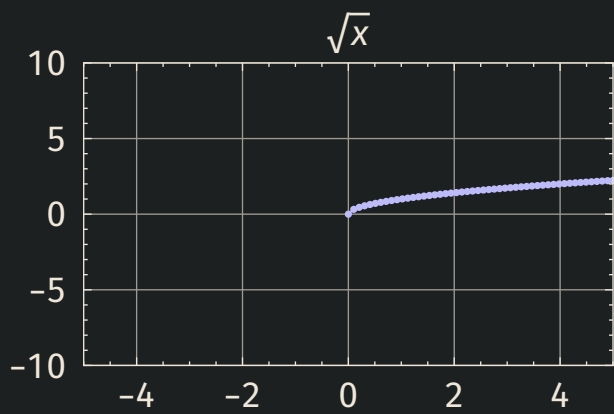


- $f(x + c) \rightarrow$ Shifts the graph c units left

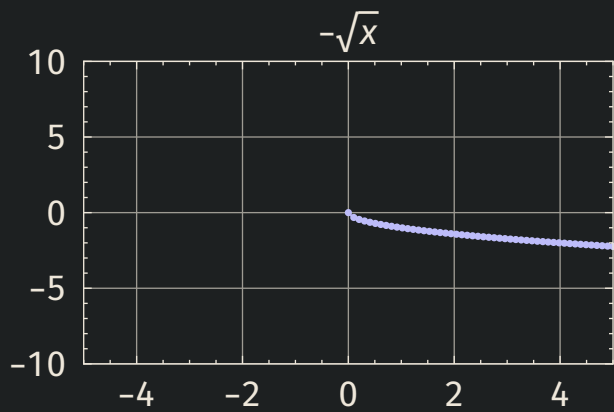


1.2. Reflections

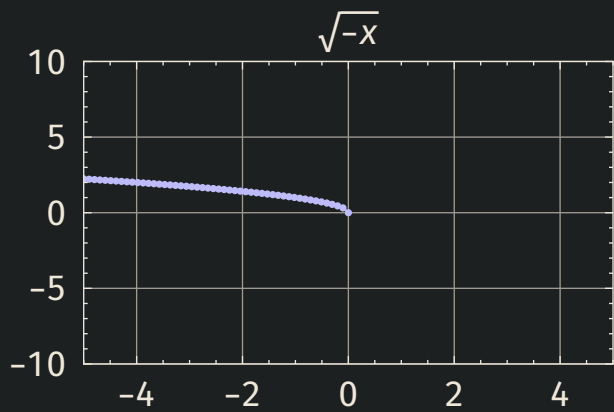
- $f(x) = \sqrt{x} \rightarrow$ Parent Function



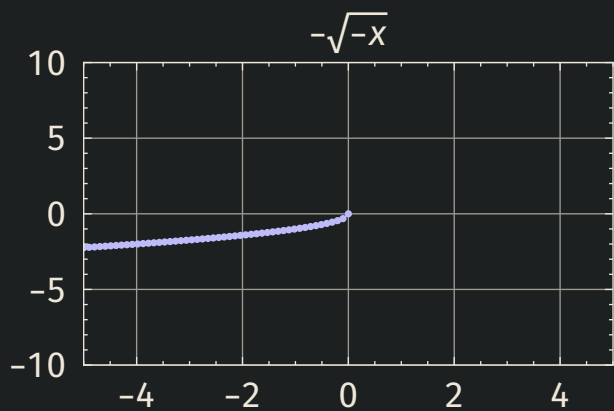
- $-f(x) \rightarrow$ Flips graph over the x-axis



- $f(-x) \rightarrow$ Flips graph over the y-axis

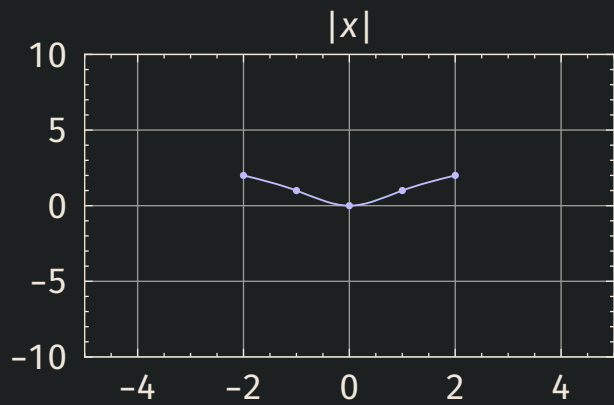


- $-f(-x) \rightarrow$ 180° rotation over the origin

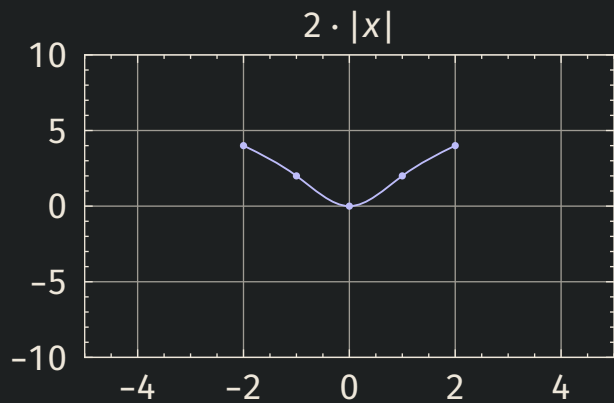


1.3. Dilations

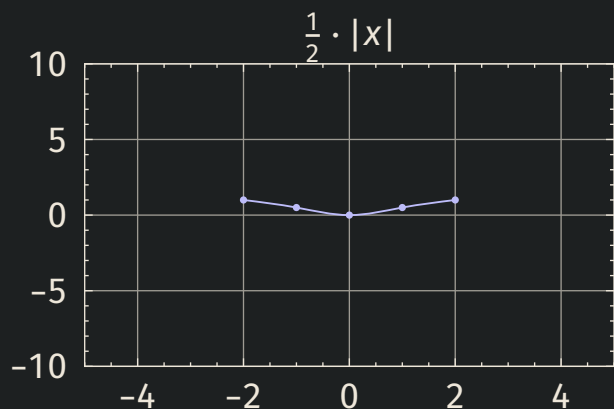
- $f(x) = |x| \rightarrow$ Parent Function



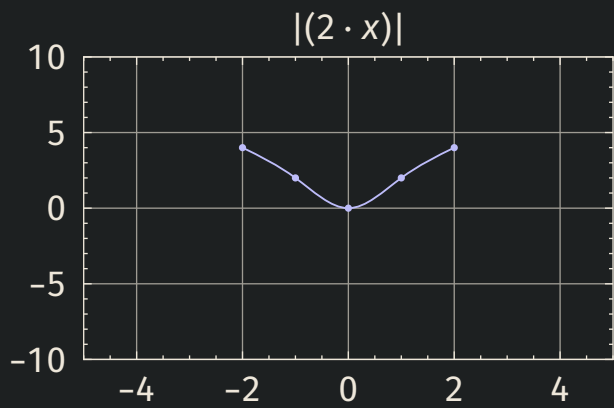
- $a \cdot f(x) \rightarrow a > 1$, Graph is stretched vertically



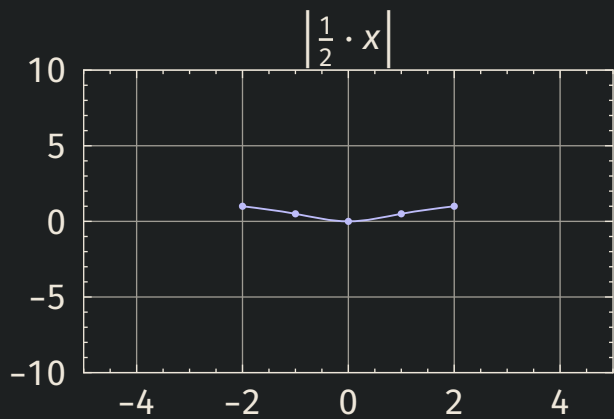
- $a \cdot f(x) \rightarrow 0 < a < 1$, Graph is shrunk vertically



- $f(a \cdot x) \rightarrow a > 1$, Graph is shrunk horizontally

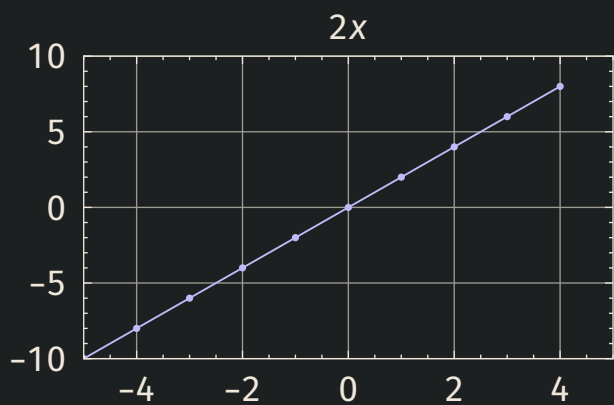


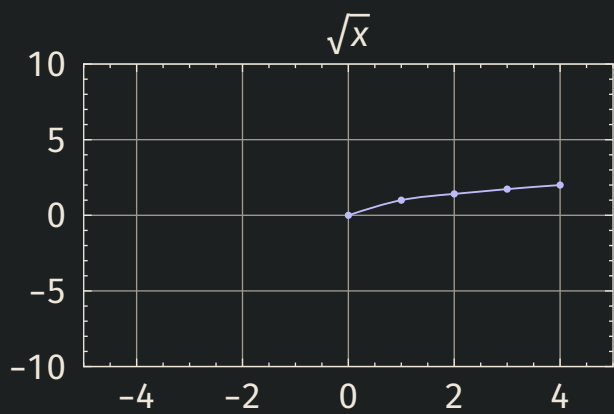
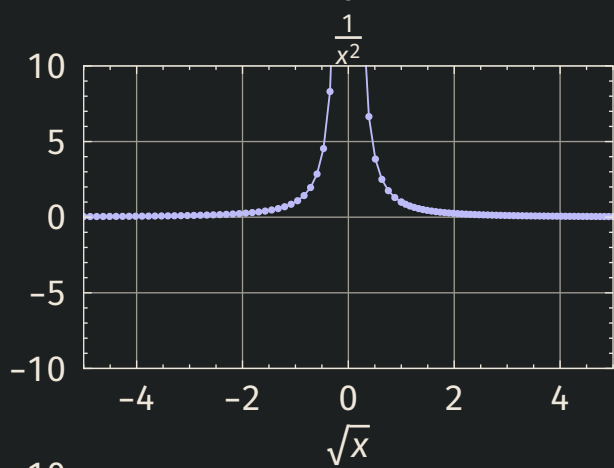
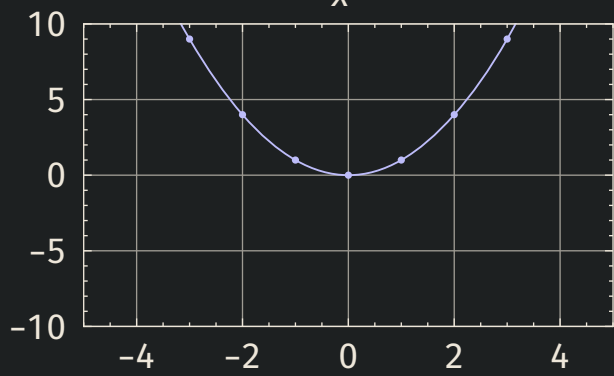
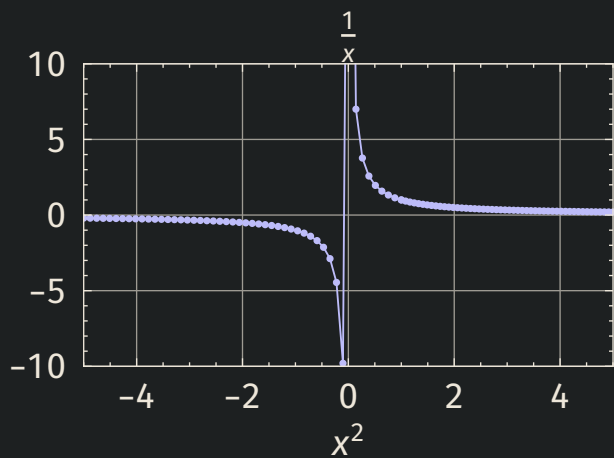
- $f(a \cdot x) \rightarrow 0 < a < 1$, Graph is stretched horizontally

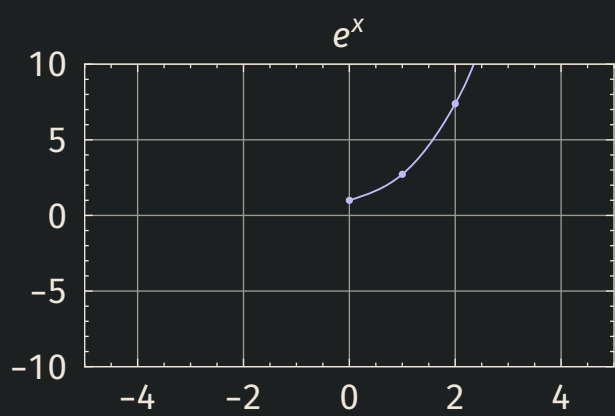
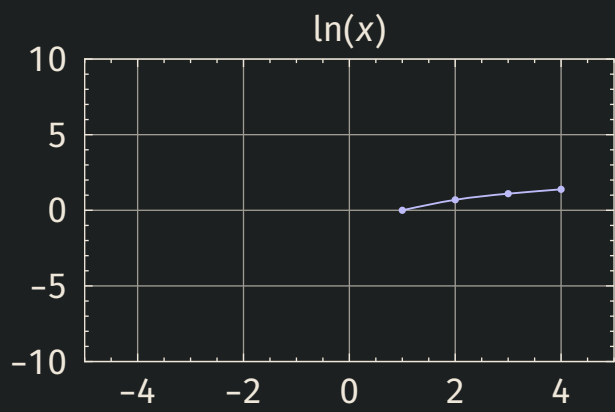
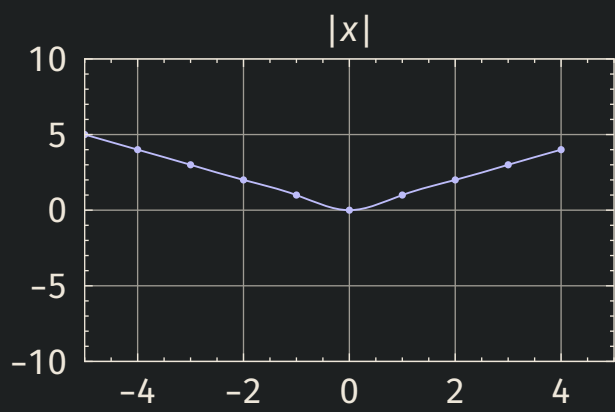


2. Graphs

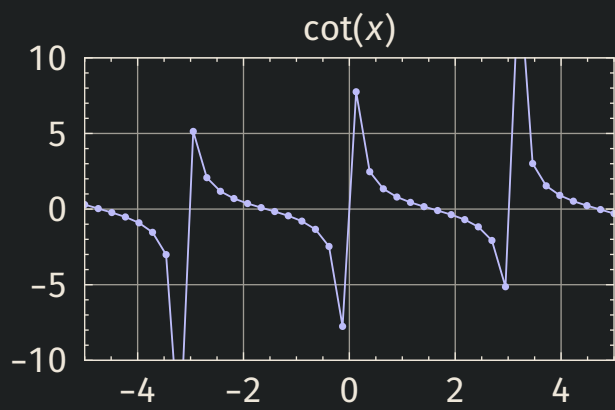
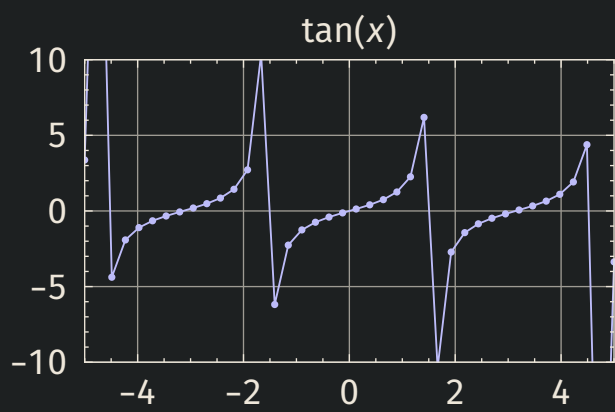
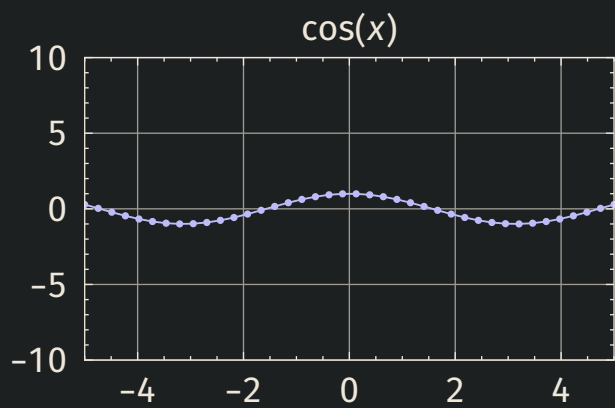
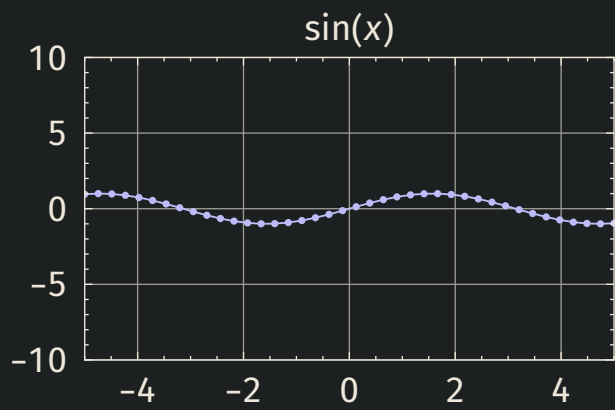
2.1. Common Graphs

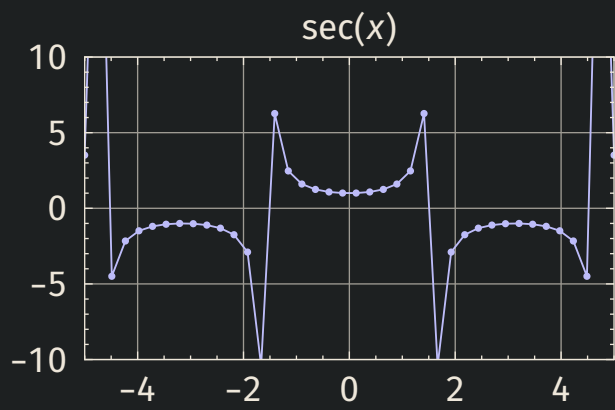
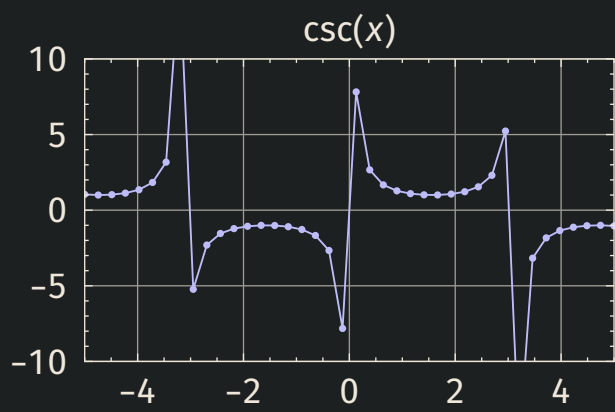




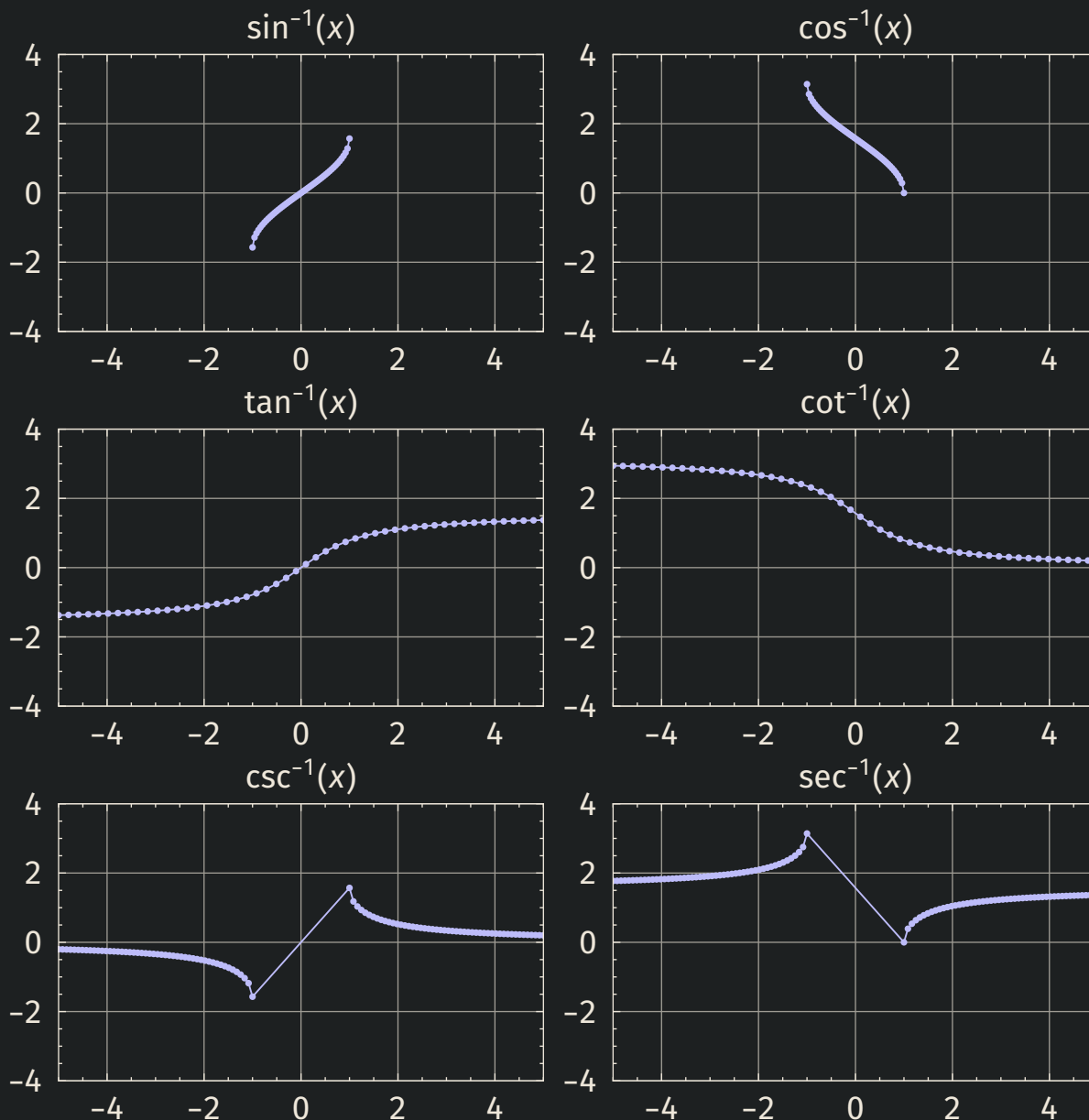


2.2. Trig Functions



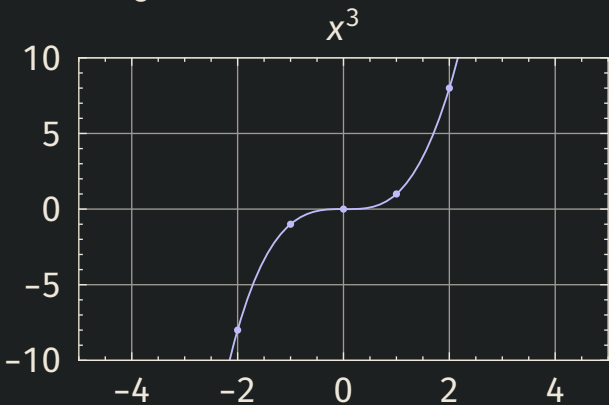
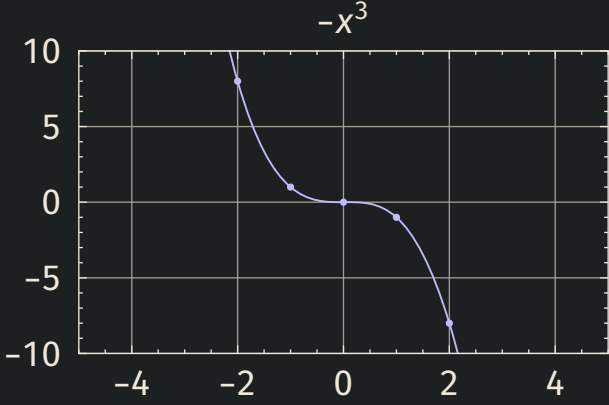
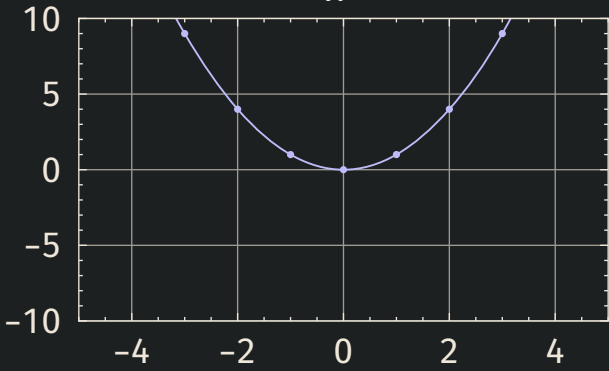


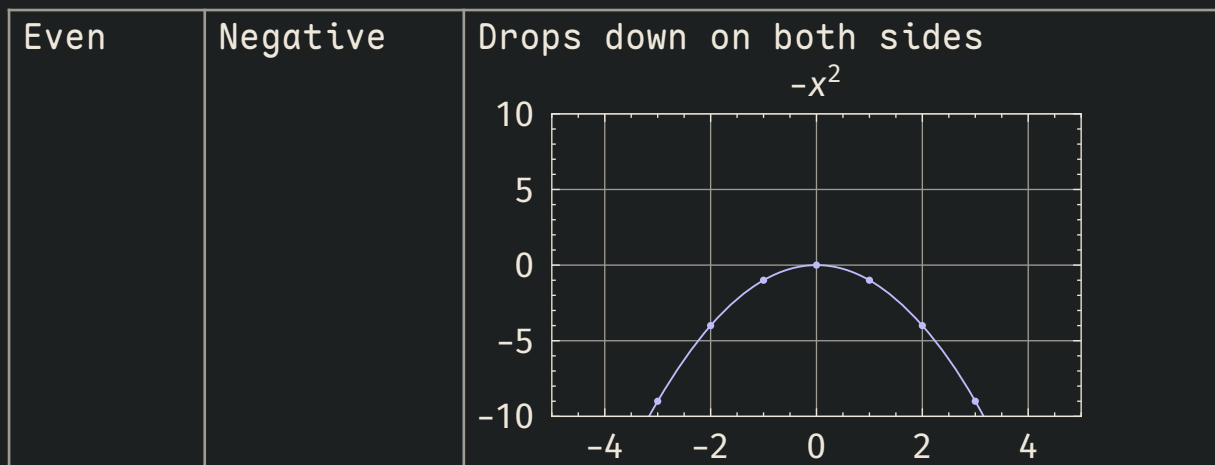
2.3. Inverse Trig Functions



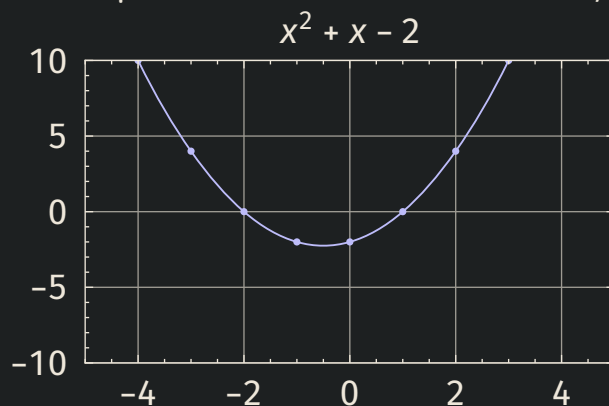
2.4. Polynomials

- **Leading Coefficient Test:** The leading coefficient test helps us evaluate the end behaviour of a function. We look at the leading term in a polynomial, that is the term with the highest exponent and we look at its exponent and coefficient. We check if the exponent is even (x^2, x^4 , etc) or odd (x^3, x^5 etc) and if the coefficient is positive or negative.

Exponent	Coefficient	Behaviour
Odd	Positive	Drops down on the left, rises on the right x^3 
Odd	Negative	Rises on the left and drops down on the right $-x^3$ 
Even	Positive	Rises on both sides x^2 

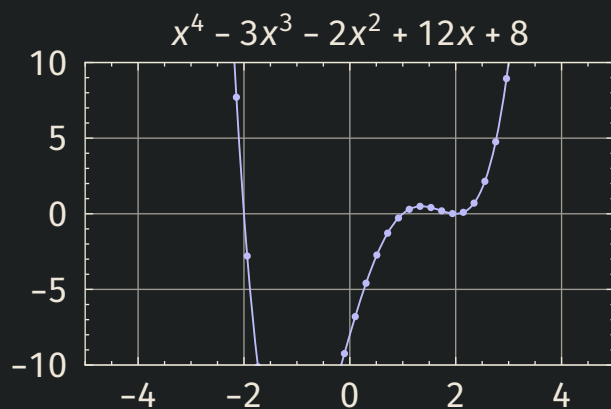


- Find the zeroes: Find the values that makes the function equal to zero. This can be understood visually as x-intercepts. To do that we solve the function for x.
- If a zero is listed odd times then the graph will cross the x-axis at that value. Ex. For the polynomial $x^2 + x - 2$ the roots are $(x-1)(x+2)$. So, this will pass the x-axis at $x=1, -2$.



- If a zero is listed twice then the graph will touch the x-axis at that value. Ex. For the polynomial $x^4 - 3x^2 - 2x^2 + 12x + 8$ the roots are $(x+2)(x-1)(x-2)^2$. So, this will pass the x-axis at $x=-2, 1$ and touch the x-axis

at $x = 2$.



Note

As the value of the exponent in the factors increases, the graph tends to get less steep near that point. Ex.

